



Training Manual



Vermicomposting



1. Introduction to Vermicomposting

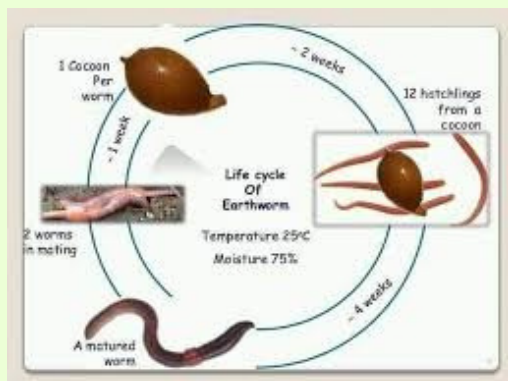
Vermicomposting is the process of using earthworms, primarily red wigglers (*Eisenia fetida*) or African nightcrawlers (*Eudrilus eugeniae*), to convert organic waste into nutrient-rich organic fertilizer (vermicompost). This eco-friendly method enhances soil health, promotes regenerative agriculture, and contributes to sustainable waste management systems.

2. Mzuri Organics Approach to Circularity

Our model integrates vermicomposting with Black Soldier Fly Larvae (BSFL) rearing and greenhouse cultivation. By combining worm castings and BSFL frass, Mzuri Organics produces potent organic biofertilizers. Innovations like Green Tags, embedding tree seeds in frass-infused biodegradable paper, extend our circularity goals to reforestation and soil regeneration.

3. Understanding the Worm Life Cycle

- Cocoon Stage: Tiny capsules containing 2–5 embryos. Hatch in 14–21 days under optimal conditions.
- Juvenile Stage: Small, colorless worms feeding on soft organics.
- Adult Worms: Mature in 6–8 weeks. Reproductive adults have a visible clitellum (band).
- Lifespan: 1–2 years under good conditions.



Key Conditions:

- Temperature: 15–26°C
- Moisture: 60–80%
- pH: 6.5–7.5
- Aeration: Critical; avoid waterlogging

4. Facility Setup and Site Considerations

- Layout: Separate zones for bedding preparation, worm beds, feeding, harvesting, and storage
- Shelter: Use simple shade structures to moderate temperature and humidity
- Flooring: Raised beds, concrete floors, or soil beds with good drainage
- Utilities: Water access, drainage systems

5. Vermicomposting Unit Operations

- Bed Construction: 1m width, any length, 20–30 cm bedding depth
- Bedding Material: wet and dry crop matter, shredded paper, fresh manure, or pre-composted organic matter
- Initial Worm Stocking Rate: 500–1000 (1–2kg) worms/m²
- Feeding Schedule: Once per cycle for single feeding system or small amounts frequently rather than large amounts occasionally for multiple feeding system.

6. Waste Collection and Pre-processing

Acceptable Waste Materials:

- Vegetable scraps, fruit peels, shredded leaves and grass, cow dung, brewers' grains

Pre-processing Steps:

- Remove plastics, metals, toxic materials
- Chop or shred bulky material
- Pre-compost high-nitrogen materials (like manure) for 7–10 days to reduce heat
- Adjust moisture to 60–70%

7. Feeding Worms and Monitoring Growth

- Feeding Frequency: Once for single feeding systems or 2–3 times per week or as consumed for multiple feeding systems
- Feeding Depth: Single feeding systems 20–30cm; multiple feeding system 5–8 cm to maintain aeration
- Moisture Monitoring: Keep bedding damp (feel like a wrung-out sponge)
- Growth Monitoring: Check worm health weekly — active, fat worms indicate good conditions

8. Vermicompost and Worm Harvesting

Harvesting Vermicompost:

- Signs: Dark, crumbly, earthy-smelling material
- Methods:
 - Light migration method (worms burrow away from light)
 - Feed migration method (worms move to fresh feed pile)
- Sieving: Use a 2–3 mm mesh to separate worms from finished compost

Worm Harvesting:

- Collect excess worms for new beds or sales

9. Post-treatment of Products

- Vermicompost:
 - Pack in breathable bags
 - Store away from sunlight and rain
- Vermiliquid:
- Worms:
 - Sold as live stock for growers
 - Used to expand in-house production
 - Used as fish feed

10. Biofertilizer and Soil Health Applications

- Vermicompost:
 - Seedling starter medium
 - Soil amendment (apply 1–2 tons per acre)
 - Compost tea brewing
- Synergistic Use: Combine vermicompost with BSFL residue for enhanced nutrient profiles

11. Hygiene, Monitoring, and Data Collection

- Clean all tools and surfaces regularly
- Keep daily records: moisture, temperature, feed input, harvesting dates
- PPE: Gloves, boots, masks
- Monitor for pests (ants, mites, etc.)

12. Worker Safety and Protective Measures

- Use gloves, waterproof boots, and protective overalls
- Handwashing stations at strategic points
- Proper lifting techniques for heavy material
- Training on health and safety

13. Appendices

Appendix A: Standard Operating Procedures (SOPs)

Appendix B: Daily Checklist Templates

Appendix C: Pictures

APPENDIX A: STANDARD OPERATING PROCEDURES

SOP 1: Preparing the Worm Bed

Objective:

Create an ideal, safe, and comfortable bedding environment for the worms to thrive and begin the vermicomposting process effectively.

Materials Needed:

- Chopped dry organic matter (leaves, maize stalk/cobs, grass cuttings, cardboard paper)
- Chopped wet organic matter (banana stem, tithornia and lantana leaves, green grass, vegetable and fruit scraps)
- Fresh cow dung
- Watering can
- Spade
- Fork jembe/jembe
- Moisture meter
- pH meter

Procedure:

1. Select a clean, shaded, and well-drained area for setting up the worm bed.
2. Prepare bedding material by mixing dry (brown) matter and green matter with manure in a 1:1 ratio (25%dry; 25%green; 50% manure)
3. Moisten the bedding material until it feels like a wrung-out sponge (approximately 60–80% moisture).
4. Place the bedding material in the bed to a depth of 20–30 cm.
5. Turn the substrate mixture every two days until the 7th day. Leave it for 3 more days and on the 10th day, sprinkle with water but do not turn.
6. Introduce worms gently onto the bedding surface on the 10th day.
7. Cover the bed with a breathable material like gunny bags or shade nets to maintain humidity or with a polythene sheet, leaving the edges open/uncovered.

Checklist:

- Bedding moist but not soggy
- pH between 6.5 and 7.5
- No sharp or toxic objects in bedding
- Worms visibly active within 24 hours

SOP 2: Feeding the Worms (domestic models)

Objective:

Maintain optimal worm nutrition to enhance vermicomposting efficiency and worm multiplication.

Materials Needed:

- Chopped organic feedstock (vegetable scraps, fruit peels, manure)
- Feed weighing scale
- Watering can
- Gloves

Procedure:

1. Pre-compost fresh feedstock for 2–3 days to reduce heat and fermentation risk.
2. Chop feed materials into small pieces (<2 cm) to hasten decomposition.
3. Weigh the feedstock to ensure consistency and avoid overfeeding.
4. Spread feedstock evenly in a thin layer over the bedding.
5. Cover the feed with a thin layer of moist bedding material to prevent pests.
6. Monitor feed consumption and replenish when 70–80% of the previous feed is consumed.

Checklist:

- Feedstock properly pre-composted
- Pieces chopped to appropriate size
- Evenly spread and lightly covered
- No foul odor or fermenting smell

SOP 3: Monitoring and Maintenance

Objective:

Ensure the vermicomposting environment remains within the optimal range for worm health and compost production.

Materials Needed:

- Moisture meter
- pH meter
- Thermometer
- Observation sheet/logbook

Procedure:

1. Check moisture levels weekly using a moisture meter; adjust with water as needed.
2. Measure bed temperature twice weekly to ensure it remains between 18–24°C.
3. Test pH monthly; it should remain between 6.0 and 7.5.
4. Visually inspect worm activity weekly; worms should be shiny, fat, and actively moving.
5. Check for mating activity and presence of worm cocoons/eggs (oval cream, brown or copper colored 2cm in diameter). Presence of eggs indicate good adaptation to their new environment.
6. Maintain a record of all measurements and observations.

Checklist:

- Moisture 60–80%
- Temperature 18–30°C
- pH 6.0–7.5
- Active worm movement observed
- Mating observed
- Eggs observed
- No foul smells or pest infestations

SOP 4: Harvesting Vermicompost**Objective:**

Safely and effectively harvest mature vermicompost while minimizing worm loss.

Materials Needed:

- Sieve (2–3 mm coffee wire mesh)
- Buckets/basins or trays
- Gloves

Procedure:

1. For domestic units with continuous feeding, cease feeding worms 7–10 days before the intended harvest.
2. Uncover the bed and allow the worms to migrate downward or sideways by using a bright light or fresh feed attractant.
3. Carefully collect the top layers of vermicompost, leaving most worms behind.

4. Sieve the harvested material to separate undecomposed particles and cocoons.
5. Store finished vermicompost in breathable bags away from direct sunlight.

Checklist:

- Compost dark, crumbly, and earthy-smelling
- Majority of worms separated successfully
- Sieved material free of large particles
- Stored properly to maintain quality

• **SOP 5: Troubleshooting Problems**

• **Objective:**

- Quickly identify and resolve common problems to maintain system health and productivity.

• **Common Problems and Solutions:**

Problem	Likely Cause	Immediate Action
Bad Smell	Overfeeding, anaerobic conditions	Remove excess food, aerate bedding
Worms Escaping	Dryness, high temperatures, acidity	Moisten bedding, adjust pH, cool environment
Pests (flies, ants)	Food exposed or dry bed	Cover food properly, increase moisture
Bedding too hot	Fresh feed generating heat	Pre-compost feed longer, remove excess feed



Mzuri
ORGANICS
NATURE'S WAY

Checklist:

- Regularly inspect and address any issues
- Maintain optimal conditions consistently
- Record incidents and corrective actions

APPENDIX B: CHECKLISTS

1. Weekly Facility Inspection Checklist

- Moisture check
- pH test
- Temperature reading
- Visual inspection for worms – Active / Sluggish / Absent
- Inspection of cocoon presence – Present/Absent
- Sifting of harvested compost – Kg
- Presence of Pests – None / Few / Many

APPENDIX B: CHECKLISTS

2. Weekly Production Summary Template

Date:/...../.....

Total vermicompost produced: _____ kg

Total vermiliquid generated: _____ kg

Product sales: _____ KES

Supervisor Name:

Sign:

APPENDIX B: CHECKLISTS

3. Monthly Production Summary Template

Date:/...../.....

- Total vermicompost produced: _____ kg
- Total vermiliquid generated: _____ kg
- Product sales: _____ KES
- Equipment maintenance conducted: [Yes/No]

Supervisor Name: Sign:

APPENDIX C: PICTURES



VermiFrass Compost



Vermicelli with red wigglers



Vermi-unit set up



Different stages of
the red wiggler



Waste
processing/crushing



Worker safety and
health/PPE



Product marketing/exhibition



Integration of bsfl and vermicomposting



Product packaging



PRODUCTS



Coriander planted with vermifrac fertilizer.



Tomatoes and kales planted with VermiFrass





Live red wigglers



blended
vermicompost
testing



Community
Engagement



Investor
engagement

THE End

Supported by;

